



Project Proposal Report

Scheduler: A Time Management Mobile Application

Submitted By: Momin Hosan Emon (242001322e)

Project Type	Mobile Productivity Application
Platform	Cross-Platform Mobile
Prepared For	Project Proposal
Status	Concept Stage / Not Yet Implemented

This report defines the proposed scope, feature set, and implementation outline for a time management mobile application designed to help users plan tasks, track productivity, run focus sessions, and review performance statistics.

1. Project Overview

The proposed application is a personal productivity and scheduling tool intended to organize daily tasks, support structured focus sessions, and visualize productivity progress through charts and statistics.

The user interface will indicate a task-centered workflow with a home dashboard, calendar-based task browsing, completed and canceled task history, focus timer controls, and a dedicated analytics area for goal tracking.

The project will be developed from the ground up, meaning all screens, state management, task operations, timer logic, visual analytics, and theme support will be implemented as part of the initial product build.

2. Problem Statement & Market Opportunity

Current digital productivity tools suffer from feature fragmentation. Users are frequently forced to navigate between multiple distinct applications to manage their schedules: one for calendar appointments, another for daily to-do lists, and a third for tracking focused work blocks (such as Pomodoro timers).

This fragmentation introduces a high cognitive load, causing friction that often leads to abandoned habits. Furthermore, existing tools rarely track missed or canceled commitments accurately, leaving users without actionable data on where their schedules fall apart.

3. Project Objectives

Provide a clear and efficient system for creating, viewing, updating, and completing tasks.

Reduce distraction during work sessions by offering a guided focus timer with configurable session and break durations.

Help users monitor productivity through daily goals, task completion indicators, and historical trends.

Offer both light and dark interfaces to improve usability in different environments.

Ensure the design remains usable on small screens and different device sizes.

4. Proposed Functional Scope

The initial release is planned to include task management, calendar navigation, task status handling, focus session control, visual progress analytics, and preference-based appearance settings.

Each major screen in the UI represents a distinct module and will be implemented as a connected workflow rather than isolated pages.

The proposed scope emphasizes local productivity features first; external integrations, cloud synchronization, and advanced collaboration tools may be considered later as optional enhancements.

5. Proposed Feature Modules

Module	Planned Implementation
Home Dashboard	A home page that summarizes the user's day, shows a completion/cancellation chart, lists the current tasks and all tasks scheduled for the day, and provides quick access to task creation, theme toggling, and the focus session screen.
Calendar View	A calendar-based task browser that allows the user to move through dates, inspect tasks for a chosen day, and switch between calendar styles such as monthly and compact multi-week views.
Task Management	Full task lifecycle support, including creation, editing, deletion, status updates, and completion marking. Tasks are shown with status labels such as Due, Completed, and Canceled.
Task History / Old Tasks	An archive view that stores previously completed and canceled tasks so users can review past activity without cluttering the active task list.
Focus Session Setup	A configuration dialog where the user can define focus duration and break duration before starting a timed work session.
Focus Timer Screen	A live session interface displaying remaining time, start/pause/resume controls, stop controls, background audio access, and a summary area for total focus time.
Focus Statistics	An analytics screen that presents daily goal progress, session-type distribution, and recent trend graphs to help the user evaluate productivity over time.
Background Focus Oriented Music for Focus Sessions	Users can play relaxing background sound or “lofi focus music” from the app during focus sessions.
Goal Management	A simple goal-editing control for setting a positive daily focus target, enabling the app to show how much time has been completed and how much remains.
Theme Support	A global appearance setting with light and dark modes so the application can remain comfortable in both bright and low-light usage conditions.
Small Screen Optimization	Responsive layouts and compact spacing rules to keep the interface functional on smaller devices while preserving readability and touch usability.

6. Proposed User Flow

1. The user opens the home dashboard and reviews current tasks, progress charts, and quick actions.
2. The user can create a new task, move into the calendar screen to inspect a specific date, or open the task history area to review older work.
3. When ready to focus, the user sets a session duration and break duration, then starts the timer.
4. During the session, the app tracks elapsed focus time, provides playback and stop controls, and allows the user to monitor the session visually.
5. Users can play relaxing background sound or “lofi focus music” from the app during focus sessions.
6. After a session is complete, the user can open the focus statistics screen to review progress against daily goals and assess recent productivity patterns.

7. Non-Functional Requirements

- Usability: the interface must remain clean, readable, and quick to navigate.
- Responsiveness: layouts must adapt to different screen sizes without clipping or overlap.
- Reliability: task and timer states must remain accurate during navigation and screen changes.
- Performance: charts and lists should render smoothly even when the user has many tasks or historical entries.
- Maintainability: the project should use a modular structure so that task management, timer logic, and analytics can be maintained independently.

8. Recommended Technical Direction

A cross-platform implementation is recommended so the same codebase can serve Android and iOS devices.

The system should be organized into separate modules for task data, timer state, analytics, and presentation to support clean development and easier future expansion.

Persistent local storage is recommended for tasks, completed history, and user preferences, while the architecture should remain ready for cloud synchronization if that feature is added later.

Charting components, dialog handling, and audio playback should be implemented as reusable UI and service layers rather than hard-coded inside individual screens.

9. Expected Deliverables

- A fully designed and implemented mobile application prototype.
- A functional task scheduling system with calendar navigation and status tracking.
- A focus session system with timer controls and productivity statistics.
- A light and dark theme experience with responsive layouts.
- A polished user interface suitable for demonstration, academic submission, or portfolio presentation.

10. Conclusion

The proposed Scheduler application is a comprehensive time management solution that combines task planning, date-based organization, focus-session control, and productivity analytics in a unified mobile experience. A well-structured product concept with strong practical value, and implementing it from scratch would result in a polished productivity tool suitable for personal use and academic demonstration.